Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
MA080G	T102	2025-08-20
Kursnamn	Matematik GR (B), Kryptografi	
Provnamn	Skriftlig tentamen	
Ort	Sundsvall	
Termin		
Ämne		

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Exam – Cryptography (MA080G)

Mittuniversitetet
Department: IMD
Responsible: Aurélio Menegon

Date: August 20, 2025
Time: 5 hours
No books, notes or symbolic calculators allowed.
Calculator make/model must be stated on the cover page.

Instructions

Answer each subquestion on a separate line. Complete solutions must be provided – reasoning and calculations must be clear and not overly brief. **ALL ANSWERS MUST BE JUSTIFIED.** Grades are based on the achievement of the course objectives.

Grading scale: A (22p), B (18p), C (14p), D (10p), E (9p)

Problems

1. **Vigenère Cipher (4p)**

   (a) (1p) A given Vigenère ciphertext contains 504 letters, and it is known that the length of the key used for the encryption is 5. Give an estimation for the Index of Coincidence using Friedman's method.

   Hint: Recall the formula

   $$k \approx \frac{0.065 - \frac{1}{26}}{I_C - \frac{1}{26}}$$

   (b) (1p) Determine the key, assuming English plaintext and using Table 1, which shows the frequency of each letter in the ciphertext grouped by position modulo 5.

   Note: The table below groups letter frequencies by key position modulo 5. Each column corresponds to a different character of the Vigenère key.

   (c) (2p) The first part of the ciphertext is:

   `VISHM HXLSI APCKM AVMUB XEWW`

   Determine the corresponding part of the plaintext, assuming the ciphertext above was encrypted with the key found in item (b).

Letter	Pos 0	Pos 1	Pos 2	Pos 3	Pos 4
A	7	0	0	7	5
B	10	1	0	0	14
C	0	4	10	0	0
D	1	3	0	6	0
E	1	6	0	3	4
F	1	5	3	3	4
G	10	1	4	9	6
H	9	0	7	4	6
I	3	6	6	0	5
J	0	6	4	9	1
K	5	7	0	9	6
L	6	1	8	9	7
M	8	1	7	4	7
N	2	0	4	0	2
O	0	0	3	0	1
P	0	2	2	0	0
Q	0	0	1	1	0
R	3	13	0	1	1
S	0	1	5	10	0
T	13	5	0	0	10
U	0	5	12	8	0
V	1	12	2	4	2
W	3	2	6	8	1
X	12	5	5	0	15
Y	4	4	12	4	1
Z	2	11	0	2	3

Table 1: Letter frequency by position modulo 5

2. **Modular Arithmetic and Group Theory (5p)**

   (a) (1p) List all units in $\mathbb{Z}/15\mathbb{Z}$.

   (b) (1p) Is 2 a unit in $\mathbb{Z}/15\mathbb{Z}$?

   (c) (2p) List all generators of $(\mathbb{Z}/10\mathbb{Z}, +)$.

   (d) (1p) What is the order of 3 in $\mathbb{Z}/10\mathbb{Z}$?

3. **Euler and Fermat Theorems (4p)**

   (a) (1p) Compute $\phi(72)$.

   (b) (1p) Use Fermat's Little Theorem to compute $3^{100} \mod 17$.

   (c) (1p) True or False: For all integers a such that $\gcd(a, n) = 1$, the following holds:

   $$a^{\phi(n)} \equiv 1 \mod n$$

   (d) (1p) Does $2^{22} \equiv 1 \mod 23$?

4. **Diffie-Hellman (4p)** Let $p = 23, g = 5$. Alice: $a = 6$, Bob: $b = 15$

(a) (1p) Compute Alice's public key.

(b) (1p) Compute Bob's public key.

(c) (1p) Compute shared secret using Alice's private key and Bob's public key.

(d) (1p) Is the shared secret the same when using Bob's private key and Alice's public key?

5. **ElGamal Encryption and Decryption (3p)**

   Let $p = 107$, $g = 2$, and suppose Alice has private key $a = 23$.

   (a) (1p) Compute Alice's public key h.

   (b) (2p) Suppose Bob sends the ciphertext $c = 21$. Decrypt the message.

6. **RSA and Pollard's ρ Algorithm (4p)** $n = 391, e = 3, f(x) = x^2 + 2x + 1 \mod 391$

   (a) (1p) Find a nontrivial factor of 391 using Pollard's ρ algorithm.

   (b) (1p) Compute $\phi(n)$ using your factor.

   (c) (1p) Find the private key d, such that $ed \equiv 1 \mod \phi(n)$.

   (d) (1p) If $c = 81$, decrypt using your key d.